

The Reflexive-Possessive Rule in Mongolian as Binding Principle A and Its Implications on English¹

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1. Introduction

In Mongolian, the suffix *-aa*, notated as RX in this paper,² is attached to a non-nominative element and expresses three types of relations between that element and a local subject: anaphoric relation, possessive relation and situational relation (Anisman 2010, Guntsetseg 2011, Janhunen 2012, Kullmann and Tserenpil 2015). This is known as “Ерөнхийлөн Хамаатуулах Нөхцөл” in Mongolian grammar, which literally means “universally relating condition/principle/rule”. This paper argues that this, which translates as “the Reflexive-Possessive Rule” (RPR), is a special type of Binding Principle A by showing that the properties of RX, the hallmark of RPR, resemble those of Principle A. RX and a reflexive marker such as *self* resist a nominative host. Both RX-marked elements and *self* anaphors are licensed/bound by a local subject. RX is blocked by switch reference and *self* anaphors are not subject to rebinding. It is demonstrated that the nature of Principle A is valuation of phi-features, which entails the evoking of a reflexive feature, in a local domain. The relevant features sit on respective D heads, which are bare phrases per se, and undergo clustering. A consequence of the proposed analysis is that *own* in English is a possessive reflexive pronoun (anaphor) and that sentences such as *John loves his pictures*, when coreferentiality is assigned to the subject and the pronoun, involve an implicit *own* and therefore are subject to Principle A, not to Principle B, with no violation of the latter.

2. Outlining Reflexive-Possessive Rule in Mongolian

The hallmark of RPR is RX. RX can be attached to either nominals or clauses excluding those with nominative case. In this paper, we are particularly concerned with accusative marked objects. Objects with RX can be divided into four types: object anaphor, possessive DP, DP containing a relative clause, and object clause. We first discuss object anaphors. As shown below, only the simplex anaphor *өөр* can be used as an object, which is attached by RX. A complex form consisting of a personal pronoun and *өөр* is disallowed.

- (1) Baatar өөр-иг-**өө** шүүмжил-sen.
Baatar-NOM self-ACC-**RX** criticize-PST³

‘Baatar criticized himself.’

- (2) *Baatar ter/tüün öör-ig-**öö** šüümjil-sen.
 Baatar-NOM 3SG self-ACC-**RX** criticize-PST
 ‘Baatar criticized himself.’

Possessive DPs marked by RX display the following properties. **First**, the anaphor *öör-in* as the possessor within the DP may or may not be overt. **Second**, *öör-in*, whether overt or covert, must be coindexed with a local subject. **Third**, when *öör-in* is overt, RX is present on either it or the possessed noun; when *öör-in* is covert, RX is present on the noun. These properties are illustrated below.

- (3) Bat (öör-in) üzeg-ø-**ee**⁴ mart-san.
 Bat-NOM self-GEN pen-ACC-**RX** forget-PST
 ‘Bat forgot his own pen.’
- (4) Bat öör-in-**öö** üzeg-ig mart-san.
 Bat-NOM self-GEN-**RX** pen-ACC forget-PST
 ‘Bat forgot his own pen.’

An RX-marked noun can act as the possessor in objects, where the noun must be interpreted as a possessum of the subject, as shown in (5). However, RX must not be present on the head noun of the object phrase, as shown in (6).

- (5) Bi naiz-in-**aa** zahia-ig unš-san.
 1st-NOM friend-GEN-**RX** letter-ACC read-PST
 ‘I read my friend’s letter.’ (Kullmann and Tserenpil 2015: 110)
- (6) *Bi naiz-in zahia-ig-**aa** unš-san.
 1st-NOM friend-GEN letter-ACC-**RX** read-PST
 ‘I read my friend’s letter.’

The properties of RX-marked possessive DPs are summarized in (7).

(7) Distributional paradigm of RX in object NPs

	NOM	GEN	ACC
	NP2 _i	<i>öör_i-in</i> -RX	NP1 _j
	NP2 _i	(<i>öör_i-in</i>)	NP1 _j -RX
	NP2 _i	NP3 _k -in-RX	NP1 _j
*	NP2 _i	NP3 _k	NP1 _j -RX

We now discuss RX-marked NP objects containing an object relative clause. With RX, the relative subject is obligatorily genitive-marked and coreferential with the matrix subject. That is, the relative

subject is the genitive anaphor *öör-in*, which may be either overt or covert, as in (8). When it is overt, *öör-in* can host RX, as in (9). However, it is mostly absent when there is another DP, for example, the ablative *bagš-aas* ‘from teacher’ in the relative clause in (10).

- (8) Bi (öör-in) sur-san hičeel(-ig)-**ee** mart-san.
 1st-NOM self-GEN learn-PST lesson-ACC-**RX** forget-PST
 ‘I forgot the lesson that I learned.’
- (9) *Bi öör-in-**öö** sur-san hičeel-ig mart-san.
 1st-NOM self-GEN-**RX** learn-PST lesson-ACC forget-PST
 ‘I forgot the lesson that I learned.’
- (10) Bi bagš-aas-aa sur-san hičeel(-ig)-**ee** mart-san.
 1st-NOM teacher-ABL-RX learn-PST lesson-ACC-**RX** forget-PST
 ‘I forgot the lesson that I learned from my teacher.’ (Bai and Cao 2024)

If the relative subject is not coreferential with the matrix subject but is possessed by it, then RX is attached to it, as in (11). RX cannot be present on the noun modified by the relative clause, as in (12). The noun modified by the relative clause, which is possessed by the matrix subject, can host RX in the case of subject relative clauses, as in (13).

- (11) Bi bagš-in-**aa** zaa-san hičeel-ig mart-san.
 1st-NOM teacher-GEN-**RX** teach-PST lesson-ACC forget-PST
 ‘I forgot the lesson that my teacher taught.’
- (12) *Bi bagš-in zaa-san hičeel(-ig)-**ee** mart-san.
 1st-NOM teacher-GEN teach-PST lesson-ACC-**RX** forget-PST
 ‘I forgot the lesson that my teacher taught.’ (Bai and Cao 2024)
- (13) Bi hičeel zaa-san bagš(-ig)-**aa** mart-san.
 1st-NOM lesson teach-PST teacher-ACC-**RX** forget-PST
 ‘I forgot my teacher who taught a lesson (to me).’

These properties are summarized in (14).

(14) Distributional paradigm of RX in object NPs with (object) relative clauses

	NOM	GEN	ACC
	NP2 _i	<i>öör_i-in</i> -RX	NP1 _j
	NP2 _i	(<i>öör_i-in</i>)	NP1 _j -RX
	NP2 _i	NP3 _k -in-RX	NP1 _j
*	NP2 _i	NP3 _k	NP1 _j -RX

In the final case in which RX is used in object clauses, similar distributional properties are observed

with RX. The embedded subject *öör-in*, the genitive anaphor coindexed with the matrix subject, may or may not be overt, as exemplified in (15). When it is not overt, RX is hosted by the embedded verb, as exemplified in (16).

- (15) Bagš (öör-in) buruud-san-**aa** meder-sen.
 teacher-NOM self-GEN go wrong-PST-**RX** admit-PST
 ‘The teacher admitted that he was wrong.’
- (16) Bagš öör-in-**öö** buruud-san-ig meder-sen.
 teacher-NOM self-GEN-**RX** go wrong-PST-ACC admit-PST
 ‘The teacher admitted that he was wrong.’

If the embedded subject is not coindexed with the matrix subject but is possessed by it, RX is present on it, but not on the verb, as shown below.

- (17) Bi bagš-in-**aa** buruud-san-ig med-sen.
 1st-NOM teacher-GEN-**RX** go wrong-PST-ACC know-PST
 ‘I realized that my teacher was wrong.’
- (18) *Bi bagš-in buruud-san-**aa** med-sen.
 1st-NOM teacher-GEN go wrong-PST-**RX** know-PST
 ‘I realized that my teacher was wrong.’

These properties are summarized in (19).

(19) Distributional paradigm of RX in object clauses

	NOM	GEN	ACC
	NP1 _i	<i>öör_i-in</i> -RX	V
	NP1 _i	(<i>öör_i-in</i>)	V-RX
	NP1 _i	NP2 _j -RX	V
*	NP1 _i	NP2 _j	V-RX

We now arrive at the following conclusion. RX requires coreferentiality between the subject and the object, between the subject and the possessor contained in the object, or between the matrix subject and the subject of the embedded clause (including a relative clause and an object clause). When coreferentiality fails, RX must not be present. That is, RX is incompatible with switch reference (SR) of subjects. Let us formulate this as follows.

- (20) RX requires the coreferentiality between X, X being a subject, and Y, Y being
- an object, or
 - a possessor, or

- c. a subject (other than X), and
- d. c-commanded by X in a local domain.

As indicated by the examples discussed above, the smallest clause that contains both X and Y serves to be a local domain. That is, Y must not be licensed by a non-local subject. In (21) and (22), the RX-marked nouns are obligatorily interpreted as possessed by the embedded subject, not by the matrix subject.

- (21) Baatar Bat-in muur-(ig)-**aa** üns-sen-ig har-san.
 Baatar-NOM Bat-GEN cat-ACC-**RX** kiss-PST-ACC see-PST
 ‘Baatar saw that Bat kissed his cat (= Bat’s cat).’
- (22) Baatar Bat-in bagš-aas-**aa** zeel-sen nom-ig-ni unš-san.
 Baatar-NOM Bat-GEN teacher-ABL-**RX** borrow-PST book-ACC-PSS[3] read-PST
 ‘Baatar read the book that Bat borrowed from his teacher (= Bat’s teacher).’

As exemplified in (23), the RX-marked *öör* must refer back to the embedded subject.

- (23) Baatar_i Dorž_j-ig öör_{i/*j}-ig-**öö** šüümjil-sen-ig har-san.
 Baatar-NOM Dorž-ACC self-ACC-**RX** criticize-PST-ACC see-PST
 ‘Baatar saw that Dorž criticized himself (= Dorž).’

If *öör* refers to the matrix subject, it must not be RX-marked but PSS-marked, as exemplified in (24).

- (24) Baatar_i Dorž_j-ig öör_{i/*j}-ig-ni šüümjil-sen-ig med-ne.
 Baatar-NOM Dorž-ACC self-ACC-PSS[3] criticize-PST-ACC know-PRS
 ‘Baatar knows that Dorž criticized him (= Baatar).’

Note that in these examples, the RX-marked element must be coindexed with a local subject. In summary, RPR has the following properties.

(25) Properties of RPR with RX as its hallmark:

- a. RX resists a nominative host;
- b. RX is licensed by a local subject;
- c. RX resists and is blocked by SR.

3. Reflexive-Possessive Rule as Binding Principle A

The three properties of RX discussed above resemble three properties of English *self* in a binding context: *self* is never attached to a nominative pronoun, is combined with pronouns to form anaphors that are bound by a local subject, and cannot be bound for the second time.

- (26) NOM-resisting property: *Chris_i said [_{CP} that himself_i was appealing].
 (27) Local-binding property: John made her_i love herself_i.
 (28) Rebinding-resisting property: [_{CP} Heidi_i believes [_{DP} Martha_j's description of herself_{*i/j}]].

This suggests that RPR in Mongolian is a type of Binding Principle A. The resemblance of the NOM-resisting property and that of local-binding/licensing are quite straightforward. The following description helps to clarify the resemblance between the third property of RPR (the SR-resisting property of RX) and that of Principle A (Rebinding-resisting property of *self*). In (12), for example, SR is obtained between the subject of the relative clause and the matrix subject, which leads to the failure of RPR. This is because *hičeel* 'lesson' is first associated with the local subject *bagš* 'teacher' before the merger of the matrix verb,⁵ and then it (*hičeel*) enters into an association (possession) relation again, but with the matrix subject this time. That is, RPR applies to the same item twice, leading to ungrammaticality. Similarly in the mono-clausal sentence (6), *zahia* 'letter' is possessed by *naiz* 'friend', where RX would be licensed by *naiz*. However, after the possessive DP *naiz-in zahia* 'friend's letter' is merged with the verb *unš* 'read', *zahia* will enter into a new possessive relation by virtue of being involved in the event of reading, which the subject initiates, and therefore it calls for RX again. However, one RX fails to go for two possessive relations at the same time. That is, *zahia*, the host of RX, fails to be interpreted as a proper possessum; in other words, the two distinct possessive relations involving the same possessum cannot be properly interpreted at LF. Consequently, the derivation crashes. As with this, in (23), *Martha* binds (*her* in) *herself*; therefore, there cannot be another NP, say, *Heidi*, to bind it. If *herself* is bound twice, the derivation crashes.

Given this, it is reasonable to say that Principle A with *self* can be viewed as a type of simplex dependence in the sense that in, for example, *John loves pictures of himself*, the subject *John* and *him* in the anaphor *him-self* are coreferential, where *self* is employed as a marker of the coreferentiality/reflexivity. In contrast, RPR is a complex dependence in the sense that in, for example, (3), *Bat* and the pronoun *öör-in* 'own', the genitive form of *öör* 'self', are coreferential, where RX is employed as a marker of the coreferentiality/reflexivity. Morphologically, *self* is always present on the pronoun, as in (29), whereas RX is present on either the reflexive pronoun (anaphor) *öör-in*, as in (4), or the possessed noun, as in (3).

- (29) John_i loves pictures of him_i-self.

For Principle A with *self*, the binder and the bindee occur simply as an antecedent, e.g., *John* in (29), and the accusative pronoun in an anaphor, e.g., *him* in *him-self*. In contrast, for RPR, the binder occurs as a nominative subject, e.g., *Bat* in (3) and (4), and the bindee occurs as the genitive pronoun *öör-in*, which is optionally present at PF. Most importantly, both the reflexive markers *self* and RX are attached only to non-nominative elements that resist rebinding and both are licensed by a local subject.

According to Kratzer (2009: 216), phi-features are transmitted from a local verbal head such as *v* to anaphors in a binding context. Reuland (2020), however, argues that binding is feature valuation, not feature transmission, that starts from Spec of TP. On this basis, with a detailed review of them left out,

Following Davis (2023), possessive pronouns in English spell out the fused outcome of the head D and the material in its Spec. On the basis of a *bare phrase structure* theory of labeling (Chomsky 1995a, b), in which non-projecting heads are equivalent to phrases, Davis (2023) assumes that a bare determiner D2 occupies Spec of D1. Applying this analysis to *his pictures*, we obtain (30). After (30) is built, fusion applies so that D2 and D1 are bundled as a single node, with their features clustered on it.⁶ Next, the VI rule applies to this structure. As a result of, *his* as a single morpheme spells out multiple syntactic nodes in a “portmanteau” fashion, as in (31) and (32).

- (30) Initial structure: [DP₁ D2_[phi] [D1' D1_[poss] [N pictures]]]
- (31) Fusion applies: [DP D_[1phi, poss] [N pictures]]
- (32) VI applies: [DP his [N pictures]]

(34) [TP [vP sbj_[phi] [DP ... D_[phi, poss] ... D_[ref, poss] ... [NP ...] ...] ...] ...]
↑ ↑
tüün-ne öör-in

4. Implications for English: *Own* as a reflexive possessive pronoun

(35) Baatar_i nom(-ig)-oo mart-san.

Baatar _i		öör _i -in	nom(-ig)- oo	mart-san.
Baatar _i	tüün _i -ne	öör _i -in	nom(-ig)- oo	mart-san.
Baatar	3SG-GEN	self-GEN	book- RX	forget-PST

‘Baatar forget his (own) book.’

Importantly, English has what has been called an adjectival pronoun, namely, *own*,⁷ which behaves in the very same way as *öör-in*. As shown in (36), *own* must be coreferential with the preceding possessive personal pronoun and the subject, as is the case with *öör-in*. In addition, *own* can be omitted because the possessive relation can be expressed by the genitive pronoun.

- (36) John_i loves [his_i own_i pictures].
 *John_i loves [his_i own_j pictures].
 *John_i loves [his_j own_j pictures].
 *John_i loves [his_j own_i pictures].
 *John_i loves [his_j own_k pictures].

An important difference between *öör-in* and *own* lies in the fact that *öör-in* can be present in the possessive DP, with a personal pronoun such as *tüün-ne* ‘his’ absent (at PF), whereas *own*, when present, must be preceded by a personal pronoun such as *his*. This difference is arguably attributed to the parametric fact that Mongolian is a zero-determiner language much like Japanese and Chinese, whereas English is not. Importantly, the properties of *öör-in* and *own* as instantiating Principle A are not affected by this difference. The eligible sentence in (36) can be paraphrased by *John_i loves [pictures of him_i-SELF]*, in which *of himself* is functionally equivalent to *his own*, both encoding the meanings of possession and reflexivity.

Tying this with the property shared by *own* and the possessive reflexive pronominal *öör-in*, it is reasonable to say that *one’s own* is a possessive reflexive pronominal with a complex morphology. This is a challenge to a previous claim (Truswell 2014: 226, for example) that English does not have a possessive reflexive pronominal. It then turns out that what does not exist in English as previously claimed is *oneself’s* and what does exist is a possessive reflexive pronominal, namely, *one’s own*. Notice that *one’s own* is not a lexical item stored in the lexicon but a constellation of multiple lexical items.

Consequently, (36) is subject to Principle A, not to Principle B. This accounts for why such sentences give rise to the *prima facie* violation of Principle B. Specifically, in (37), there is an implicit possessive pronoun *own*, as illustrated in (39), which is not true in (38). In (39), the complex *his own* as a whole behaves as a possessive anaphor.

- (37) John_i loves his_i pictures.
 (38) John_i loves his_j pictures.
 (39) John_i loves his_i own_i pictures.

It then follows that the so-called “lexical” ambiguity of the possessive pronouns such as *his* between a

reflexive and a pronoun (Truswell 2014: 224) is in fact a structural ambiguity, as fleshed in (40) and (41). We apply the analysis presented in section 3 to *his own* as follows.

(40) [_{CP} John_i loves [_{DP} his_j [_N pictures]]]

(41) [_{CP} John_i loves [_{DP} his_i [_{DP} own_i [_N pictures]]]]

(42) Initial structure: [_{DP1} D3_[phi] [_{D1'} D2_[ref] [_{D1'} D1_[poss] [_N pictures]]]]

Unlike the case of *his pictures*, both fusion and the VI rule apply twice throughout (42) and (43). When they apply to D3_[phi] and D1_[poss], *his* spells out their fused outcome, i.e., D_[phi, poss], as in (44), and when they apply to D2_[ref] and D1_[poss], *own* spells out their fused outcome, i.e., D_[ref, poss]. Importantly, after the application of fusion, DP becomes a layered projection, containing two D heads. The higher D takes care of personal possession, and the lower D takes care of reflexive possession.

(43) Fusion applies: [_{DP} D_[phi, poss] [_{DP} D_[ref, poss] [_N pictures]]]

(44) VI applies: [_{DP} his [_{DP} own [_N pictures]]]

Note that the features [ref] and [poss] are bundled on the lower head D, which leads to the possessive reflexive property of *own*. Valuation of [phi] takes place, and [ref] is evoked, as shown below.

(45) [_{TP} [_{vP} subj_[phi] [_{DP} ... D_[phi, poss] ... D_[ref, poss] ... [_{NP} ...] ...] ...]
↑ ↑
his own

5. Conclusion

This paper argued that RPR in Mongolian is a special kind of Principle A by showing that RPR has three important properties resembling those of Principle A. Regarding the implementation of binding, it was argued that [phi] on the D head is valued by [phi] on the subject within a local domain.

The proposed analysis has four consequences. First, anaphors are not necessarily the hallmark of binding since they can be absent at PF. Second, there are implicit possessive anaphors. Third, English *own* is a reflexive possessive pronoun (possessive anaphor), functionally equivalent to Mongolian *öör-in*, and can be absent at PF. Fourth, English possessive pronouns such as *her* are not lexically ambiguous between non-reflexive and reflexive pronouns.

It is hoped that further exploration will reveal more facts about Principle A both in Altaic languages centering on Mongolian and English-type languages on the one hand and syntactic generalizations of binding on the other hand.

NOTES

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² RX, being subject to vowel harmony, has four allophonic morphemes, *-aa*, *-ee*, *-oo* and *-öö*, which do not differ from each other syntactically and semantically.

³ The abbreviations to use in this paper include ACC: accusative, GEN: genitive, NOM: nominative,

PRS: present, PSS: possessive suffix, PST: past, and RX: reflexive-possessive suffix.

⁴ The accusative marker *-ig* is mostly not overt, without affecting the use of RX. However, for the anaphor *öör*, it must be overt.

⁵ The possessive relation signaled by RX does not necessarily entail a literal ownership; associative possession, situational possession, and alienable/inalienable possession can all be possessive relations in a broader sense.

⁶ *Fusion* here, a term of Distributed Morphology (Halle and Marantz 1993; Embick and Marantz 2008), refers to a syntactic operation that gets two (or more) nodes united into one before the application of the morphological operation VI.

⁷ In Present-Day English, *own* is often labeled “adjective”. However, it differs from stacked adjectives in many respects and displays properties of pronouns or determiners (König and Vezzosi 2008).

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